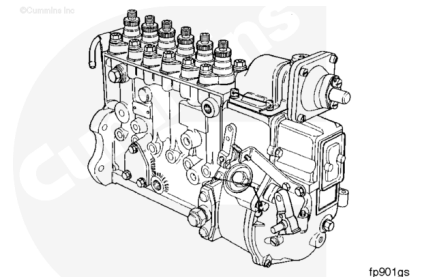


Trainings ▾

General Information

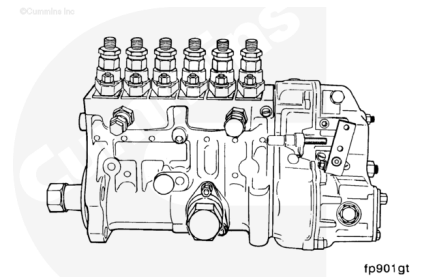
The Bosch® P7100 in-line fuel injection pump can be found on the following engine application:

- 1991 high-horsepower automotive ratings
- 1994 automotive, all 6B ratings
- 1996 emissionized high-horsepower industrial ratings.



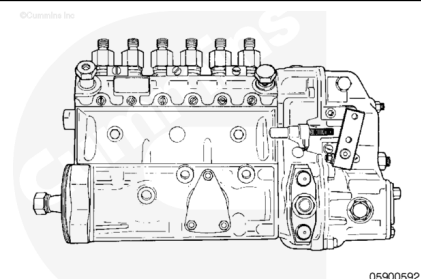
The Nippondenso EP-9 in-line fuel injection pump with the RSV governor can be found on the following engine application:

- Higher horsepower marine rating.



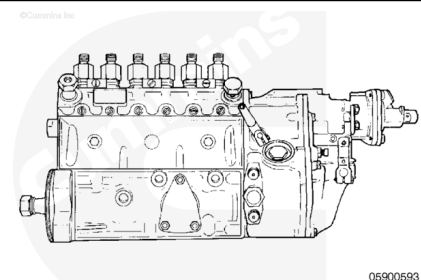
The Bosch® A-RSV in-line fuel injection pump can be found on the following engine application:

- Pre-1996 noncommissioned industrial ratings
- 1996 industrial emission ratings
- Marine
- Gensets.



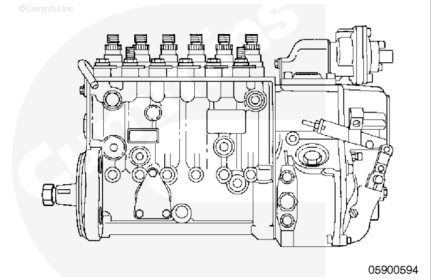
The Bosch® MW/RSV and MW/RQV in-line fuel injection pump can be found on the following engine application:

- 1996 and higher industrial emission ratings.



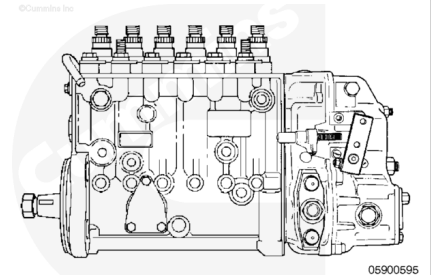
The Bosch® P3000/RQVK and P3000/RQV in-line fuel injection pump can be found on the following engine application:

- 1996 and higher high-horsepower industrial emission rating.



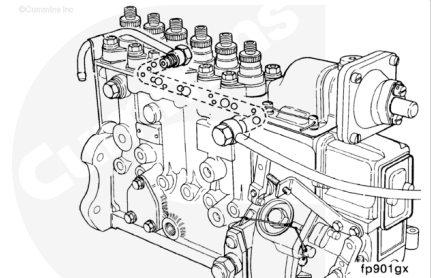
The Bosch® P3000/RSV In-line fuel injection pump can be found on the following engine application:

- 1996 and higher high-horsepower industrial emission ratings
- Marine.

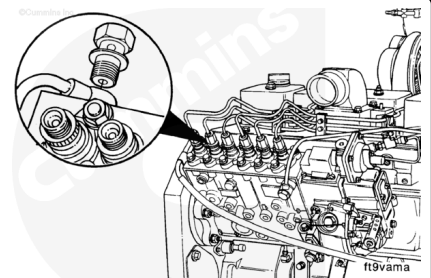


The pressure relief valve arrangement on the Bosch® P7100 fuel injection pump in the supply side of the fuel circuit creates a self-bleeding system for air introduced during replacement of the supply-side components.

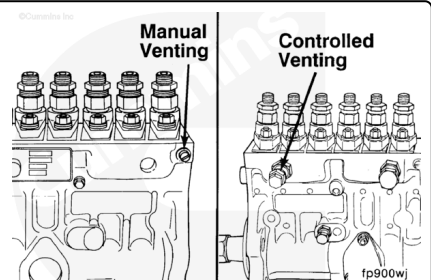
Small amounts of air can be bled from the pump by operating the hand primer on the fuel transfer pump or by cranking the engine.



The Bosch® P7100 in-line fuel injection pump has a jump-over tube to route return fuel and entrapped air from the pressure relief valve directly to the supply tank.



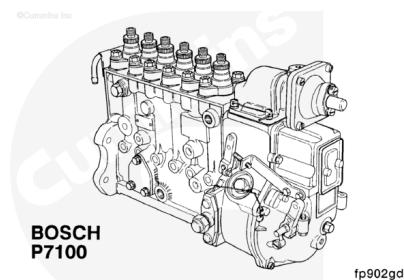
The Nippondenso EP-9 in-line fuel injection pumps will require additional venting prior to initial start-up, pump replacement, or if engine fuel runs out.



Fuel Injection Pump (In-Line Type)

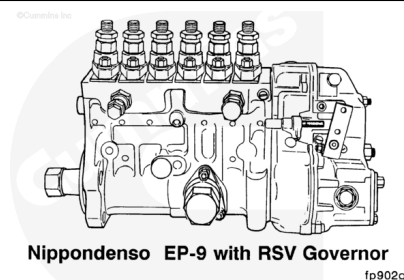
Beginning in 1991, the B Series engine used the Bosch® P7100 in-line fuel injection pump on higher horsepower automotive ratings. In 1994, all automotive 6B Series engines used the Bosch® P-7100 in-line fuel injection pump.

B Series industrial ratings and marine (after 1996) engines use the Bosch® “A” in-line fuel injection pumps.



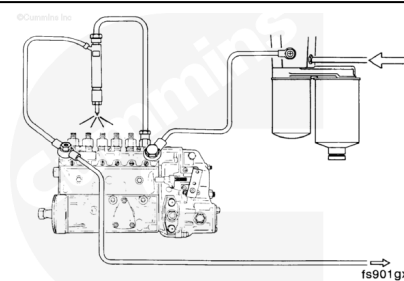
The B Series engine also uses the Nippondenso EP-9 with RSV governor on 1996 or earlier engines with a 250 and 300-horsepower marine rating.

Refer to the B Series Marine Operation and Maintenance Manual, Bulletin 3810466, for additional information.



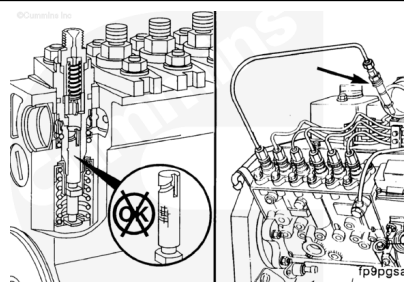
The fuel injection pump performs the three basic functions of:

1. Metering the exact amount of fuel for each injection cycle
2. Producing the high fuel pressure required for injection
3. Delivering the high-pressure metered fuel to each cylinder at the precise time.



Individual plungers are used in the pumps to develop and distribute the high pressure required for injection.

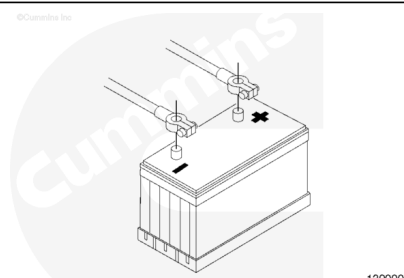
A worn or damaged plunger in the pump will affect **only** one cylinder.



Preparatory Steps

⚠ WARNING ⚠

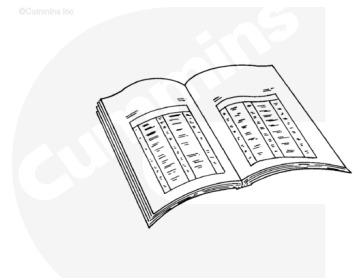
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Disconnect the batteries.

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arching equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



ck800wa

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Clean any debris from the fuel injection pump.
- Remove the fuel supply lines. Refer to Procedure 006-024 (</qs3/pubsys2/xml/en/procedures/40/40-006-024-tr.html>).
- Remove the injector supply lines to the pump. Refer to Procedure 006-051 (</qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html>).
- Remove the control linkage. See the equipment manufacturer service information.
- Remove the fuel shutoff valve. Refer to Procedure 005-043 (</qs3/pubsys2/xml/en/procedures/40/40-005-043.html>).
- Remove the air fuel control air tube. Refer to Procedure 006-001 (</qs3/pubsys2/xml/en/procedures/40/40-006-001-tr.html>).
- Disconnect wastegate turbocharger control line (if applicable).

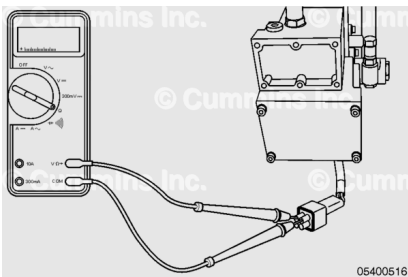
Adjust

Electronic Governor

Resistance Check

Make sure that coil is not connected before checking the electronic governor resistance.

Use the volt-ohm meter to check the coil resistance.

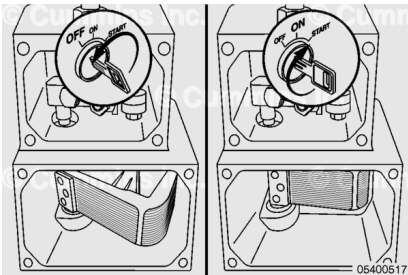


Coil Resistance	
Voltage	Resistance
24 VDC	7.2±0.2 ohms
12 VDC	1.9±0.2 ohms

Actuator Check

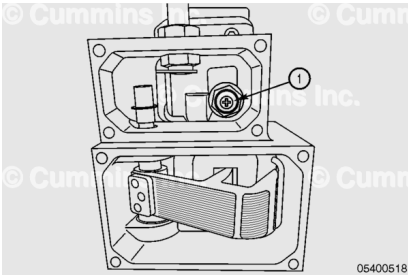
Remove the electronic governor cover.

Turn the key switch to the "ON" position.



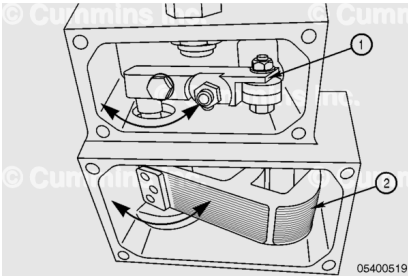
Fuel injection pump gear lever (1) flexibility checking.

When the actuator is pulling in, the gear lever **must** be stretched out and drawn out back.



Operation lever to check for flexibility.

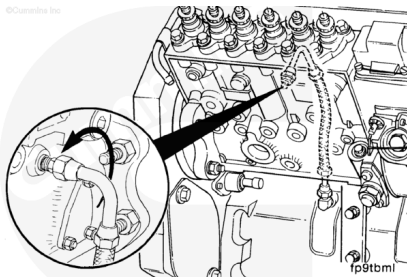
Move the actuator (1) back and forth. Lever (2) **must** move freely.



Remove

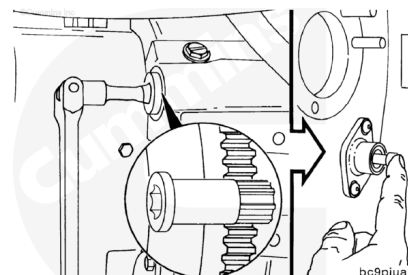
Disconnect the external oil feed line at the inboard side of the fuel injection pump (if applicable) and the main oil rifle.

Disconnect the external oil feed line at the rear of the pump or AFC latchout if applicable.

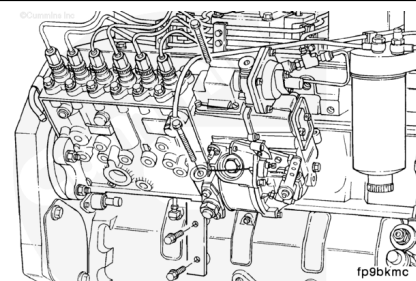


Locate top dead center for cylinder Number 1. Push the top dead center pin into the hole in the camshaft gear while slowly barring the engine.

Note : Be certain to disengage the timing pin after locating top dead center.



Remove the fuel injection pump mounting bracket, if applicable.

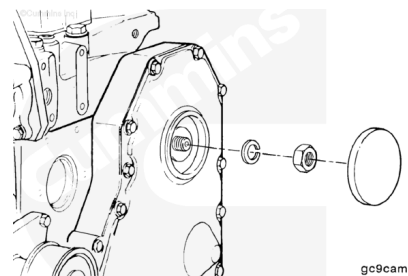


⚠ CAUTION ⚠

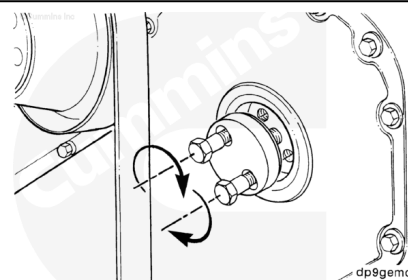
Do not drop the nut and washer. Dropping the nut and washer will result in the need to remove the front cover.

Remove the gear cover access cap.

Remove the nut and washer from the fuel injection pump shaft.

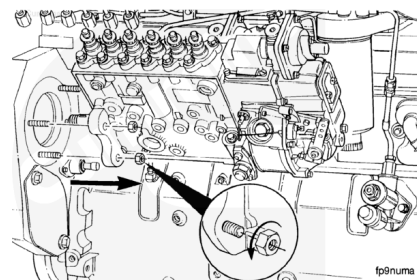


Use fuel pump gear puller, Part Number 3163381 or Part Number 3824469 with M8-1.25 x 50 capscrews, grade 8.8 or equivalent. Pull the fuel injection pump drive gear loose from the shaft.



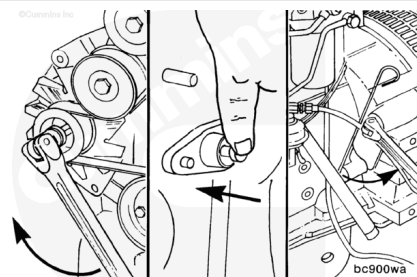
Remove the four mounting nuts.

Remove the fuel injection pump.

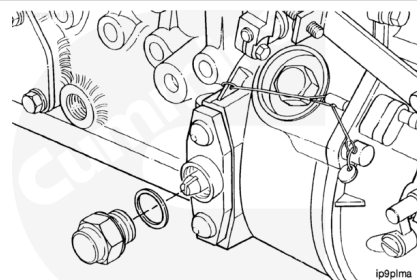


Install

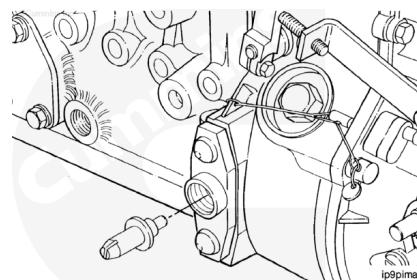
Make certain that the engine has cylinder Number 1 at top dead center.



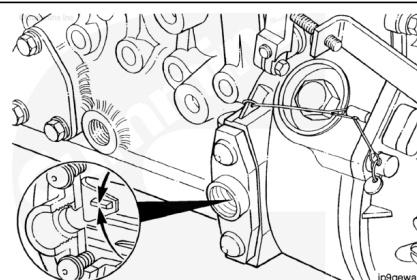
Remove the access plug.



Remove the timing pin.

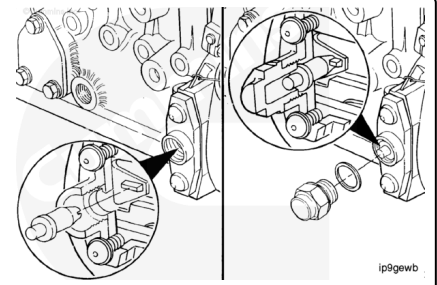


If the timing tooth is **not** aligned with the timing pin hole, rotate the fuel injection pump shaft until the timing tooth aligns.

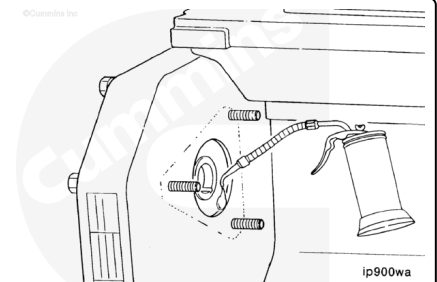


Reverse the position of the timing pin so the slot of the timing pin will fit over the timing tooth in the pump.

Install and secure the timing pin with the access plug.

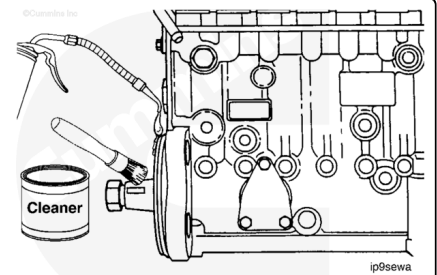


Use a 50/50 mixture of clean lubricating engine oil and STP, or equivalent, to lubricate the gear cover housing to make certain that the fuel injection pump will slide into the gear cover housing easily.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

⚠ CAUTION ⚠

The fuel injection pump drive gear inside diameter and the shaft outside diameter must be clean and dry before installing the gear. Failure to do so can result in slipped timing.

Note : Before installing the fuel pump drive gear, clean the injection pump shaft and gear tapers with residue-free cleaner, Part Number 3824510, by spraying into the gap between the shaft and the gear. Dry with compressed air.

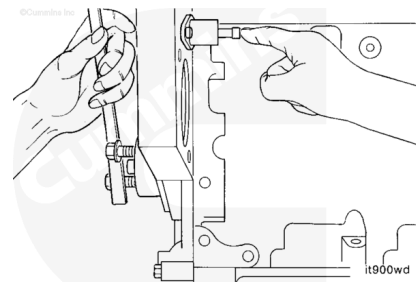
Note : The in-line fuel injection pump driveshaft has a provision for a Woodruff key: however, it is not

required. Timing mark alignment is **not** required for the in-line drive gear.

Note : Make certain that the engine has cylinder Number 1 at top dead center.

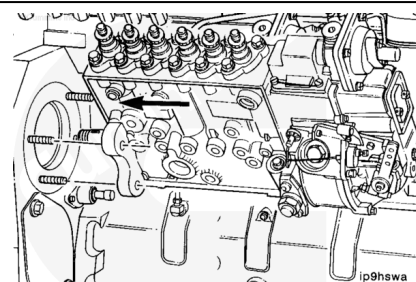
Make certain that the o-ring seals for the fill orifice and pilot are correctly installed and are **not** damaged.

Install new pilot o-ring.



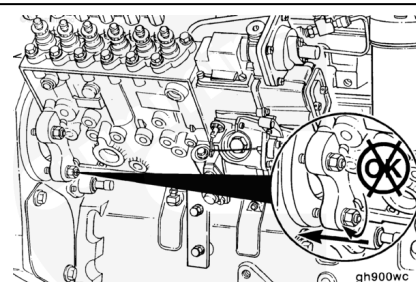
Slide the pump shaft through the drive gear and position the pump flange onto the mounting studs.

Push the pump forward until the mounting flange and o-ring are properly fitted into the gear housing bore.



⚠ CAUTION ⚠

Do not attempt to pull the pump flange into the gear housing with the mounting nuts as damage to housing can occur.

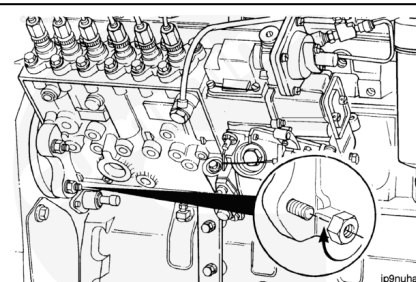


Install the mounting nuts.

Torque Value: 43 n•m [32 ft-lb]

Install the support bracket (if equipped).

Torque Value: 32 n•m [24 ft-lb]

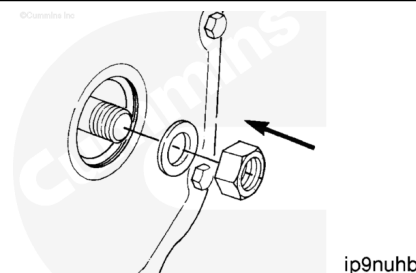


⚠ CAUTION ⚠

Do not drop the nut and washer. Dropping the nut and washer will result in the need to remove the front cover.

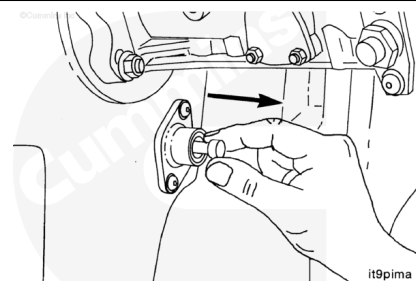
Install the retaining nut and washer.

Torque Value: 10 to 15 n•m [89 to 133 in-lb]



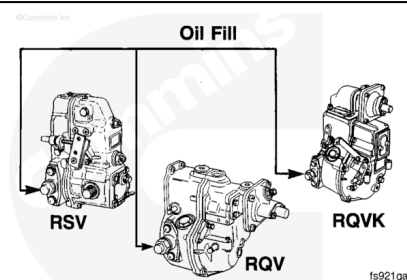
To prevent damage to the timing pins, do **not** exceed the torque value given. This is **not** the final torque value for the retaining nut.

Disengage the engine timing pin.



⚠ CAUTION ⚠

The governor housing must be prelubricated before engine operation. Failure to do so can result in premature governor wear.



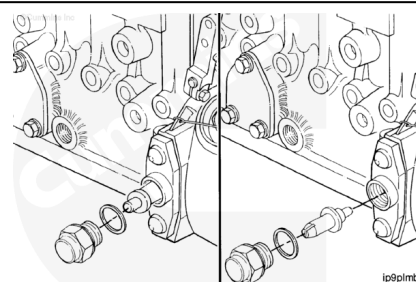
Remove the access plug.

Add the following quantity of clean lubricating engine oil:

- RSV 450 mL [0.48 qt]
- RQV 750 mL [0.79 qt]
- RQVK 750 mL [0.79 qt]

Remove the fuel injection pump timing pin plug, reverse the position of the timing pin, and install the timing pin, plug, and sealing washer.

Torque Value: 27 n•m [20 ft-lb]



Tighten the fuel injection pump drive nut.

Torque Value:

"A" Pump

1. 85 n•m [63 ft-lb]

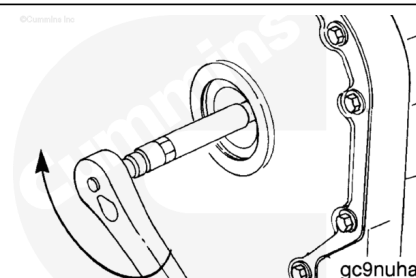
Torque Value:

P3000 and P7100

1. 195 n•m [144 ft-lb]

Torque Value:

Nippondenso



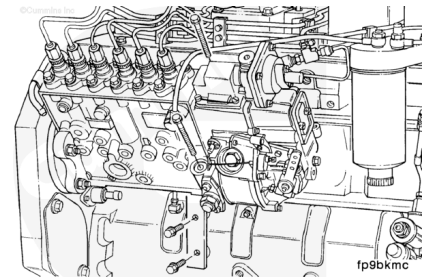
1. 123 n•m [91 ft-lb]

Install the gear cover access cap hand-tight.

Install the fuel injection pump mounting bracket capscrews.

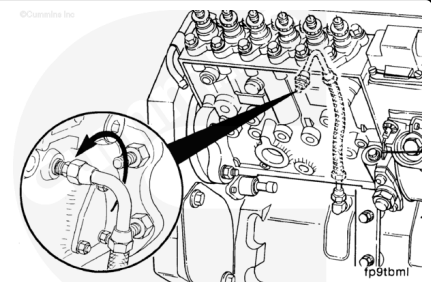
Tighten all capscrews by hand for proper alignment.

Torque Value: 24 n•m [18 ft-lb]



Connect the external oil feed line at the inboard side of the fuel injection pump (if applicable) and the main oil rifle.

Connect the external oil feed line at the rear of the pump or AFC latchout if applicable.



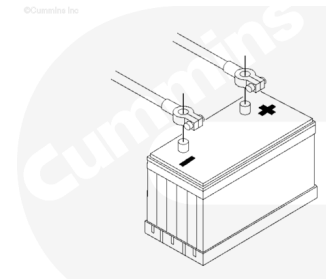
Finishing Steps

- Connect wastegate turbocharger control line (if applicable)
- Install the air fuel control air tube. Refer to Procedure 006-001 (</qs3/pubsys2/xml/en/procedures/40/40-006-001-tr.html>)
- Install the fuel shutoff valve. Refer to Procedure 005-043 (</qs3/pubsys2/xml/en/procedures/40/40-005-043.html>)
- Install the control linkage. See the equipment manufacturer service information.
- Install the injector supply lines to the pump. Refer to Procedure 006-051 (</qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html>)
- Install the fuel supply lines. Refer to Procedure 006-024 (</qs3/pubsys2/xml/en/procedures/40/40-006-024-tr.html>).



Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries
- Operate the engine and check for leaks.



Prime

Replacing the fuel supply lines, fuel filters, fuel injection pump, high-pressure fuel lines, and injectors will let air enter into the fuel system. Follow the specified procedure to bleed the air from the system.

Refer to Procedure 006-015

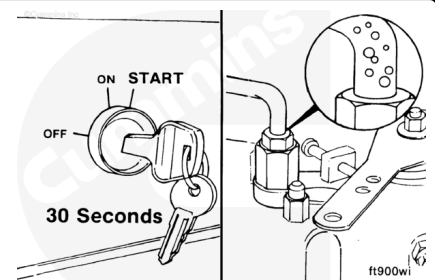
(/qs3/pubsys2/xml/en/procedures/40/40-006-015-tr.html)

Fuel Filter (Spin-On) for proper venting of the low pressure side of the fuel system.

Refer to Procedure 006-051

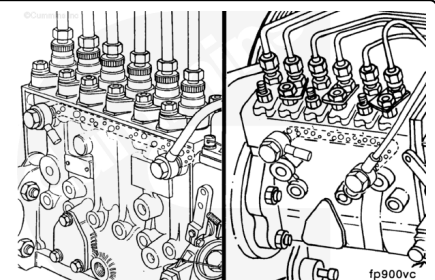
(/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)

Injector Supply Lines (High Pressure) for venting of the high pressure side of the fuel system.



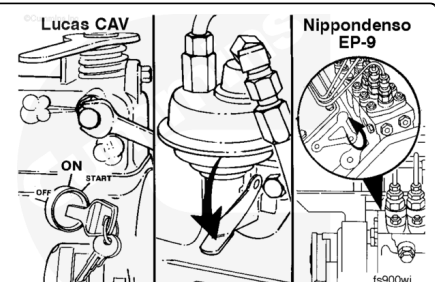
The MW, A, and P fuel injection pumps equipped with the engine-side fuel drain arrangement create a self-bleeding system for air introduced during replacement of the supply-side components.

For faster air purge, small amounts of air can be bled from the pump by operating the hand primer on the fuel transfer pump or by cranking the engine.



Injection Pumps - Venting

After priming the low pressure fuel lines (see Procedure 006-015, Fuel Filter – Spin-On Type), air/fuel can be vented from the illustrated vent locations on the Nippondenso EP-9 and the Lucas CAV fuel injection pumps. The Lucas CAV fuel injection pump requires that the fuel solenoid valve be energized before venting.



Loosen the vent screw, and operate the priming lever on the fuel transfer pump until the fuel injection pump is primed.

Tighten the vent screw.

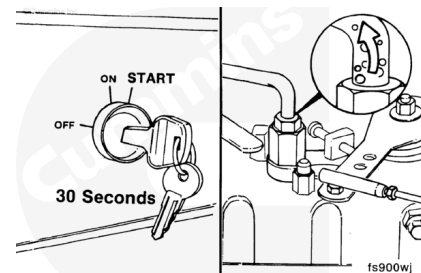
Torque Value: 9 n•m [80 in-lb]

⚠ CAUTION ⚠

It is necessary to turn the keyswitch to the ON position. Because the engine can start, be sure to follow all safety precautions. Use the normal engine starting procedure.

⚠ CAUTION ⚠

When using the starting motor to vent the system, do not engage it for more than 30 seconds, or starter damage will occur. Wait 2 minutes before starting the engine again.



Air can also be vented through the fuel drain manifold line by operating the starting motor.